

MODULAR CONTRACTS

Jason Roderick Donaldson Giorgia Piacentino Zichen Zhao

FACTS

Current theory: Contract as incentive/sharing rule: outcome $\mapsto$ transfers

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Real contracts

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Evaluated clause by clause

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Conservatism: High burden of truth to verify clause

QUESTION

Why are contracts modular, with clauses evaluated separately?

What explains loopholes, complexity, landmines, incompleteness,...?

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Model of modular contracts

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Clauses collections of atomic premises

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Premises evaluated (1 or 0) based on threshold of observed data

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Marginal unchanged $\implies$ evaluate $\mathbf{a}_j, \mathbf{a}_k$ correctly

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Marginal unchanged $\implies$ evaluate $\mathbf{a}_j, \mathbf{a}_k$ correctly, not joint $\mathbf{a}_j \wedge \mathbf{a}_k$

RESULTS

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Narrow clauses

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Extension: High thresholds $\implies$ conservatism

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Contract Φ

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Contract Φ : Set of clauses $\{\mathcal{M}_m\}_m$ evaluated & conjoined

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Objective

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Objective: Minimize $C = c_I \Pr[\llbracket \Phi \rrbracket > \llbracket \phi_{\mathcal{T}} \rrbracket] + c_{II} \Pr[\llbracket \Phi \rrbracket < \llbracket \phi_{\mathcal{T}} \rrbracket]$

MODEL: EXAMPLE: SHOE DELIVERY

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Evaluation: $\llbracket \hat{\phi} \rrbracket = 1$ if $\# \{\text{shoes/pairs}\} \geq 2$

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(Atomic) Premises: $\mathbf{a}_\ell$: left shoe $\mathbf{a}_r$: right; Data: size $i \in \{1, 2, 3\}$

Target policy: $\phi_{\mathcal{T}} = \mathbf{a}_\ell \wedge \mathbf{a}_r$ (“[deliver] pairs”: $a_\ell^i \wedge a_r^i$)

Observations = shuffled data: $\hat{\mathbf{a}}_j$ sizes shuffled w.p. e

Clause: “right shoe,” “left shoe,” “pairs”

Evaluation: $\llbracket \hat{\phi} \rrbracket = 1$ if $\# \{\text{shoes/pairs}\} \geq 2$

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MODEL: EXAMPLE: COVENANTS

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Premises

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DEFINITIONS: PREMISE- AND TARGET-BASED

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$\Phi^{\mathcal{T}}$ called target-based if there is one clause s.t. $\mathcal{M} = \bigcup_{j \in \mathcal{T}} \{\mathbf{a}_j\}$

BENCHMARKS: $I = 1$ AND $e = 0$

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Each premise is just a truth value in $\{0, 1\}$, so $\llbracket \phi \rrbracket = \phi$

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Evaluation reduces to logic; grouping of premises irrelevant

BENCHMARK: $e = 0$

No shuffling, so $\hat{\mathbf{a}}_j = \mathbf{a}_j$ for all j

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$\Rightarrow$ target-based contract implements target policy: $\llbracket \Phi^{\mathcal{T}} \rrbracket = \llbracket \phi_{\mathcal{T}} \rrbracket$

LOOPHOLE AND SHUFFLE RISK SETS

DEFINITIONS: PREMISE- AND TARGET-BASED

Loophole states LH: Data realizations s.t. $\llbracket \Phi^{\mathcal{A}} \rrbracket \neq \llbracket \Phi^{\mathcal{T}} \rrbracket$ even if $e = 0$

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Loophole states LH: Data realizations s.t. $\llbracket \Phi^{\mathcal{A}} \rrbracket \neq \llbracket \Phi^{\mathcal{T}} \rrbracket$ even if $e = 0$

Shuffle risk states SR: Data realizations s.t. $\Pr\left[\llbracket \Phi^{\mathcal{A}} \rrbracket \neq \llbracket \Phi^{\mathcal{T}} \rrbracket \mid \{\mathbf{a}_j\}_j\right] > 0$

EXAMPLE

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Two premises with three data each and $\tau = 2$:

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i
1
2
3

EXAMPLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$
1	1
2	1
3	0

EXAMPLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$
1	1
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# of 1s $\geq \tau$	

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i	$\mathbf{a}_j$
1	1
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EXAMPLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$	$\mathbf{a}_k$
1	1	1
2	1	0
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Premise-based: $\llbracket \Phi^{\mathcal{A}} \rrbracket$

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Two premises with three data each and $\tau = 2$:

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1	1	1
2	1	0
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# of 1s $\geq \tau$	1	1

Premise-based: $[\Phi^A] = [\mathbf{a}_j] \wedge [\mathbf{a}_k] = [110] \wedge [101]$

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Two premises with three data each and $\tau = 2$:

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Target-based:

EXAMPLE

Two premises with three data each and $\tau = 2$:

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1	1	1	1
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Target-based: $\llbracket \Phi^T \rrbracket = \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket$

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Premise-based: $[\Phi^A] = [\mathbf{a}_j] \wedge [\mathbf{a}_k] = [110] \wedge [101] = 1 \wedge 1 = 1$

Target-based: $[\Phi^T] = [\mathbf{a}_j \wedge \mathbf{a}_k] = [110 \wedge 101] = [100]$

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i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
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Loophole: Evaluation & conjunction don't commute: $\llbracket \mathbf{a}_j \rrbracket \wedge \llbracket \mathbf{a}_k \rrbracket \neq \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket$

EXAMPLE: SHUFFLE RISK

Observe permuted data

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
3	0	1	0
$\# \text{ of 1s} \geq \tau$	1	1	0

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Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle)

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$	$\hat{\mathbf{a}}_j$
1	1	1	1	1
2	1	0	0	1
3	0	1	0	0
# of 1s $\geq \tau$	1	1	0	1

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Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle) ; $\hat{\mathbf{a}}_k = 110$ (shuffle of $\mathbf{a}_k$)

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$	$\hat{\mathbf{a}}_j$	$\hat{\mathbf{a}}_k$
1	1	1	1	1	1
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Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

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3	0	1	0	0	0	0
# of 1s $\geq \tau$	1	1	0	1	1	1

Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

Joint changed: $101 \rightarrow 110 \Rightarrow$ target-based evaluation flips

RESULTS

LEMMA

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK

LH $\subsetneq$ SR

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

LH $\Rightarrow$ SR

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LH $\Rightarrow$ SR: Loophole states: Premise-based accepts, target-based rejects

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Premise based accepts if $\min_j \sum_i a_j^i \geq \tau$

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SR $\nRightarrow$ LH: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$

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SR $\nRightarrow$ LH: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$: $\llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket = \llbracket 110 \rrbracket = 1$ if $\tau = 2$ (not LH)

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But shuffle could be $\hat{\mathbf{a}}_j = 110$, $\hat{\mathbf{a}}_k = 101$

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But shuffle could be $\hat{\mathbf{a}}_j = 110$, $\hat{\mathbf{a}}_k = 101$: $\llbracket \hat{\mathbf{a}}_j \wedge \hat{\mathbf{a}}_k \rrbracket = \llbracket 100 \rrbracket = 0$ (yes SR)

R1: LOOPHOLES

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With $e > 0$, if $\Pr[\text{SR} \setminus \text{LH}]$ large, then $\Phi^{\mathcal{A}} \succ \Phi^{\mathcal{T}}$

R1: LOOPHOLES: INTUITION

Shuffled data ($e > 0$): Evaluation errors for all but not only loopholes

Accept loopholes to reduce shuffling error

LOOPHOLES IN PRACTICE: J. CREW

Distressed J. Crew wanted to free collateral

Credit agreement ostensibly prohibited moving it to unrestricted sub, but...

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I.e. loophole: $\llbracket \mathbf{a}_j \rrbracket \wedge \llbracket \mathbf{a}_k \rrbracket \neq \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket$

LOOPHOLES IN THE LITERATURE

Katz 10: Many loopholes openly exploited after much litigation

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Show why loopholes persist $e > 0$

R2: COMPLEXITY RISK

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If $c_I \Pr[\text{LH}]$ sufficiently high, $\Phi^{\mathcal{T}} \succ \Phi^{\mathcal{A}}$

R2: COMPLEXITY RISK: INTUITION

If loopholes likely or costly use complex target-based contract (long clauses)

Cost: With interacting premises, shuffle risk $\Rightarrow$ likely evaluated with error

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Empirically: Complexity leads to errors in taxes and regulator compliance
OECD, Benzarti 20, Zwick 21, Behn et al 22...

R3: REDUNDANCY

DEFINITION: REDUNDANCY

$\Phi^{\mathcal{R}}$ called redundant if $j \in \mathcal{M}_m \cap \mathcal{M}_{m'}$ for some clauses $j \in \mathcal{A}$, $m \neq m'$

R3: REDUNDANCY

If $e > 0$ not too high and $c_1 \text{Pr}[\text{LH}]$ high enough, both $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{T}}$ & $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{A}}$

R3: REDUNDANCY: INTUITION

Benefit w.r.t. target based: Fewer false positives

Repeated clause $\Rightarrow$ require multiple shuffling errors

Benefit w.r.t. premise-based: Fewer loopholes

Keeps joint information

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Keeps joint information

Cost: False negatives: Shuffles in any clause can cause rejection

R4: INCOMPLETENESS

DEFINITION: INCOMPLETENESS

$\Phi^{\mathcal{I}}$ called incomplete w.r.t. $\mathcal{T}$ if $j \notin \bigcup \mathcal{M}_m$ for some $j \in \mathcal{T}$

R4: INCOMPLETENESS

If e and c_{Π} high and $\Pr [\llbracket \Phi^{\mathcal{I}} \rrbracket > \llbracket \phi^{\mathcal{T}} \rrbracket]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

R4: INCOMPLETENESS: INTUITION

Long clauses induce shuffling risk $\Rightarrow$ omit some premises to avoid it

Akin to mismeasurement in Holmström–Milgrom:

Decrease power on distortionary tasks

R5: CONSERVATISM

EXTENSION

Clause-dependent thresholds $\tau_{\mathcal{M}}$

DEFINITION: CONSERVATISM

Let $\tau_{\mathcal{T}}$ be threshold of target

$\Phi^{\mathcal{C}}$ is conservative if $\tau_{\mathcal{M}_m} > \tau_{\mathcal{T}}$ for every m

R5: CONSERVATISM

Define $\text{ACC}_\tau(\Phi) := \{ \{ \mathbf{a}_j \}_j : \llbracket \Phi \rrbracket = 1 \}$

If $\Pr [\llbracket \phi_{\mathcal{T}} \rrbracket = 1 \mid \text{ACC}_\tau(\Phi^{\mathcal{A}}) \setminus \text{ACC}_{\tau+1}(\Phi^{\mathcal{A}})] < \frac{c_{\text{I}}}{c_{\text{I}} + c_{\text{II}}}$, then $\Phi^{\mathcal{C}} \succ \Phi^{\mathcal{A}}$

R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

Feature: Relies on only marginals $\Rightarrow$ not subject to shuffle risk

R5: CONSERVATISM: APPLICATIONS

Law: “beyond reasonable doubt,” “reasonable suspicion,” “probable cause”
In re Winship, 397 U.S. 358 (1970), Terry v. Ohio, 392 U.S. 1 (1968)

Accounting & auditing: High standard for good news Basu 97; Watts 03

E.g. losses recognized faster than gains

CONCLUSION

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Model of modular contracts in formal language s.t. shuffled data

Loopholes

Errors of complexity

Landmines

Incompleteness

Conservatism

MODULAR CONTRACTS